

**1ST
EDITION**
EFFECTIVE 17TH SEP, 2025

**TEST CASES FOR ANNEXURE:
CANCER CARE & ONCOLOGY
MANAGEMENT – NABH DIGITAL
HEALTH STANDARDS (HIS & EMR)**

FORWARD

NABH, a constituent board of the Quality Council of India, has been at the forefront of ensuring quality and excellence in healthcare services for the past 19 years. Over the years, NABH standards have transformed healthcare delivery, raising awareness among healthcare workers, patients, and their families about their rights and responsibilities. Dedicated to the SDG-3-Target 2030 and Viksit Bharat 2047, we aim to ensure healthy lives and promote well-being for all ages, with a special focus on patient safety and empowerment.

The evolving healthcare landscape, driven by technological advancements, has made Hospital Information Systems (HIS) and Electronic Medical Records (EMR) crucial for efficient patient data management and enhanced communication among healthcare providers.

With the mission to establish an accreditation program that sets a benchmark for the progress of the healthcare industry, fostering a culture of quality and safety in healthcare, I am delighted to introduce NABH's latest initiative—the first edition of standards for HIS and EMR systems in India.

These pioneering standards are aligned with NABH core standards and the NABH Digital Health Standards for Hospitals, marking a transformative shift in digital healthcare. This alignment places NABH in a globally unique position, where a single entity oversees all three standards, developed with consistent rigour. By upholding the highest standards in digital health, we are laying the foundation for a more efficient, transparent, and patient-centered healthcare ecosystem in India."

The new standards include Objective Elements (OEs) structured to be assessed across four levels: Core, Commitment, Achievement, and Excellence- mirroring the structure of our Digital Health Standards for Hospitals. Developed by a committee of technology specialists, NABH technical experts, and leading medical practitioner, these standards cover critical clinical and administrative workflows, data security, revenue management, interoperability, and scalability.

NABH acknowledges the commendable work being done by NHA and the significant role of the ABDM platform in promoting interoperability. We also value NHA's effort in evaluating and certifying Digital Health products for ABDM interoperability and ensuring necessary security certifications (including WASA testing). Hence NABH standards for HIS/EMR systems mandate that all products must be evaluated and approved by NHA for ABDM and security requirements, prior to applying for NABH certification. This alignment strengthens the collaboration between NHA and NABH ensuring that robust Digital Health solutions get certified and used by hospitals across India.

In developing the NABH certification for HIS/EMR systems, we have also drawn inspiration from global standards in Digital Health, as well as best practices in software development and security. We have also engaged extensively with industry experts, who have made valuable contributions to these standards. We are truly excited about the potential of NABH certification for HIS/EMR systems to accelerate the adoption of Digital Health in India, by driving standardization in product requirements and making it easier for hospitals to procure high-quality HIS/EMR products.

We recognize that developing these standards is an ongoing journey, and we welcome feedback from Digital Health companies, hospitals, and other organizations to continuously refine the standards. Additionally, we are working on the testing and certification approach for HIS/EMR systems including the selection of NABH-approved software testing agencies.

As we embark on this journey, I urge all clinicians, healthcare organizations, HIS and EMR system companies, other stakeholders, and policymakers for their support in the adoption of these standards. Together, we can elevate healthcare quality, promote patient-centric care, and contribute to a healthier India where every individual benefits from cutting-edge digital health solutions.

I extend my best wishes to all products adopting these standards and applaud their commitment to quality and patient safety. Thank you for your dedication to excellence, and I look forward to seeing the positive impact of these standards on our healthcare ecosystem.

The 1st edition of NABH Digital Health Standards for HIS/ EMR Systems serves as a valuable resource for organizations committed to providing safe, effective, and patient-centered care. May this edition inspire a new era of excellence in healthcare advancing our shared goal of ensuring that every patient receives the highest standard of care.

NABH remains committed to its mission of taking Quality, Safety and Wellness to the last in the line.



Dr. Atul Mohan Kochhar
CEO, NABH

ACKNOWLEDGEMENT

I would like to express my sincere appreciation and gratitude to all the individuals and organizations who have contributed their time, knowledge, and expertise in preparing the 1st edition of NABH Standards for HIS and EMR Systems.

I would place my heartfelt thanks and deepest gratitude to Shri Jaxay Shah, Chairperson QCI, for his vision to take quality to the grassroots and permeate the idea of quality in the DNA of every citizen in every part of India.

Mr. Rizwan Koita, Chairperson NABH, has been the guiding light throughout the development of this first edition of NABH Standards for HIS and EMR Systems. I thank him for his active participation, support, and invaluable suggestions despite his busy schedule.

I sincerely thank Mr. Chakravarthy T. Kannan, Secretary General of the Quality Council of India for his guidance and continuous support in making adequate resources available for this initiative.

I extend my heartfelt thanks to all the NABH board members for their invaluable suggestions, which have greatly contributed to the enhancement of the standards and their accompanying guidebooks.

The technical committee of NABH worked relentlessly and meticulously to accommodate the best practices in HIS/EMR space, referred to innumerable academic references, and incorporated suggestions made by all the stakeholders in bringing these standards to reality.

I thank the team of PricewaterhouseCoopers Private Limited (PwC) for their contribution in developing these standards.

I extend my deepest gratitude to team of Koita Foundation for their technical inputs and unwavering commitment to achieve this milestone.

I thank all our diligent assessors, owners of the HIS/EMR system companies, and various other stakeholders who gave us extensive feedback to improve upon the standards and their exhaustive interpretation.

I express my deep appreciation to the officers at the NABH Secretariat for their tireless efforts in completing the work on time. Their dedication and teamwork have been instrumental in delivering this guidebook with such thoroughness and precision.

Dr. Atul Mohan Kochhar
CEO, NABH

Introduction to Oncology Annexure and Objective Elements Test Cases

This document contains the test cases developed for evaluating the NABH Digital Health Standards and objectives for Oncology annexure. These test cases are meticulously designed to ensure that the HIS and EMR systems meet the required performance, functionality, and compliance standards. The purpose of these test cases is to provide a structured, unified, and comprehensive approach to testing, enabling NABH, HIS and EMR vendors to maintain a uniform standard while testing different HIS and EMR products. With the help of these test cases, the HIS and EMR vendor can assess the current gaps in their systems while doing the Self-Assessment (SAT) and accomplish the testing phase successfully.

Structure of the Test Case

Each test case is divided into four key parts to ensure thorough and systematic testing.

1. **Test Case:** This section provides a brief description of the specific feature or functionality being tested. It outlines the objective of the test.
2. **Test Validation:** This section outlines the mode of testing. An objective can be validated by self-attestation (submitting relevant certification) or by manually demonstrating the steps of the test case.
3. **Pre-requisite:** Before executing the test case, certain conditions or configurations must be in place. This section lists all the necessary prerequisites, including system settings, data requirements, and any other dependencies that must be satisfied.
4. **Steps to Follow:** This section provides a detailed, step-by-step procedure to execute the test case. Each step is carefully articulated to ensure clarity and consistency, enabling testers to accurately carry out the test.
5. **Expected Outcome:** This section describes the anticipated results upon successful execution of the test case. It serves as a benchmark against which actual outcomes are compared to determine whether the test has passed or failed.

Usage Guidelines

- **Preparation:** Ensure all prerequisites are met before starting the test.
- **Execution:** Follow the steps meticulously to avoid any discrepancies.
- **Documentation:** Record the actual outcomes and compare them with the expected outcomes.
- **Reporting:** Document any deviations, bugs, or issues encountered during testing and take necessary actions.

AAC.3 : The system handles laboratory test orders and samples.

AAC.3.a : The system configures clinical and administrative workflow for laboratory management.

Test Case : Verify configuration and end-to-end workflow of laboratory management including Oncology Module.

Pre-requisite for test	Test Validation	
1. The laboratory management module must be enabled in the system. 2. Master data for laboratory tests, practitioners, and billing codes must be pre-configured. 3. User must have valid credentials with appropriate access rights for laboratory management and oncology workflow configuration. 4. Laboratory Information System (LIS) integration should be established and active. 5. EMR module should be configured for clinician authorization workflows. 6. Keep a dummy patient whose tumour board meeting is scheduled in the system.	Manual	
Steps to produce	Expected Outcome	Note/Deviation
Step 1. Navigate to the Laboratory Management module from the main dashboard. Step 2. Set up laboratory workflow configurations to enable information flow between practitioners, the lab, and billing. Step 3. Configure essential quality control parameters within the laboratory management system. Step 4. Generate specialty-specific reports such as microbiology and biochemistry to validate reporting capabilities. For the Oncology requirement Step 5. Initiate a molecular-specific sample using barcode-based tracking and validate correct entry. Step 6. Initiate EMR-enabled test requisition workflow with clinician authorization and confirm seamless data transfer to LIS. Step 7. Validate bidirectional integration by ensuring LIS updates are reflected in the EMR. Step 8. Generate structured bioinformatics reports for molecular test results, which may include (Patient & Case Details, Molecular Test Performed, Tools, Databases & References, Primary Molecular Findings, etc) and	1. Laboratory master data, workflows configurations are successfully set up. 2. Quality controls are properly configured and applied. 3. Specialty-specific laboratory reports (e.g., microbiology, biochemistry) are generated accurately. 4. Molecular-specific sample registration with barcode tracking is successfully completed. 5. EMR-enabled requisition workflows function correctly with clinician authorization.	Select Yes/No

Steps to produce	Expected Outcome	Note/Deviation
integration into the patient's EMR and longitudinal health history for dummy patient. Step 9. Verify generation of unified reports combining molecular biology findings with IHC and histopathology results with stage .(As Applicable) Step 10. Confirm that the unified report is accessible for tumor board discussions and supports development of integrated treatment plans.	6. LIS and EMR integration is seamless with real-time bidirectional updates. 7. Structured bioinformatics reports are generated and integrated into the patient's EMR and longitudinal history. 8. Unified reports combining molecular, IHC, and histopathology data are generated successfully. 9. Unified reports are accessible for tumor board use and support comprehensive treatment planning.	Select Yes/No

COP.1 : The system manages consultation services in OPD and IPD

COP.1.a : The system records and reviews initial assessment in OPD and IPD, and patient progress.

Test Case: Verify recording and review of initial assessment and patient progress in OPD and IPD, including Oncology Enhancements.

Pre-requisite for test	Test Validation
1. User must have valid credentials with roles assigned (doctor, nurse, or authorized staff). 2. Master data for patient categories (antenatal, obstetrics, paediatrics, ophthalmology, ENT, oncology, etc.) must be configured. 3. EMR must be integrated with diagnostic, radiology, and laboratory systems. 4. Oncology-specific modules (Surgical Oncology, Radiation Oncology, Chemotherapy Oncology) must be enabled and accessible. 5. Keep two dummy patients registered in the system.	Manual

Steps to produce	Expected Outcome	Note/Deviation
Step 1. Navigate to the OPD module and initiate a dummy patient encounter. Step 2. Record the initial assessment details including vital signs, medical history, physical examination findings, diagnostic tests, and previous treatments. Step 3. Save and review the recorded initial assessment and confirm its availability in the patient's medical record. Step 4. Repeat the initial assessment process for IPD patients and validate proper storage and review functionality. Step 5. Select a patient category (antenatal, obstetrics, paediatrics, ophthalmology, ENT, etc.) and capture category-specific assessment and re-assessment details. Step 6. Navigate to the patient progress monitoring section and review the ability to track trends, vitals, diagnostic results, and treatment progress over time. Step 7. Capture and confirm the primary diagnosis details, including date of diagnosis, TNM staging, and modalities used such as imaging, histo-cytopathology, and special tests. Step 8. Document initial radiotherapy assessment including scan details, patient positioning, and immobilisation information. Step 9. Capture surgery-related data including procedure details, name, extent, frozen sections, and HPR. Step 10. Record chemotherapy and radiation therapy details including cycles, intervals, dates, type, indication (adjuvant, concurrent, neoadjuvant, palliative), and treatment response or toxicity. Step 11. Update patient records with chemotherapy scheduling details including indications, vitals, laboratory results, and radiology reports/images. Step 12. Record the cause of death if it occurs during or within 30 days post-treatment, specifying the modality involved. Step 13. Validate that the identity of the user entering or updating the data is captured and logged by the system.	- Initial assessment records are successfully captured and reviewed in both OPD and IPD modules. - Category-specific assessments and re-assessments are recorded and retrievable. - Diagnostic tests, primary diagnosis, and TNM staging are accurately captured and stored. - Radiotherapy-specific details are recorded, including scans, positioning, and immobilisation. - Surgery-related data, including extent and pathology results, are stored in the system. - Chemotherapy and radiation therapy data including cycles, type, indication, and treatment responses are captured and reviewed. - Chemotherapy schedules with diagnostic results and vitals are updated and accessible. - Cause of death data is properly recorded when applicable.	Select Yes/No

Steps to produce	Expected Outcome	Note/Deviation
Step 14. Generate and review oncology reports referencing Annexure (A) Surgical Oncology, (B) Radiation Oncology, and (C) Chemotherapy Oncology modules.	10. User identity for each data entry is logged and auditable. 11. Comprehensive oncology reports are generated and accessible as per defined modules.	

COP.1 : The system manages consultation services in OPD and IPD

COP.1.b : The system provides a summary of the patient's condition, medication order and follow-up visit for the OPD visit.

Test Case : Verify summary of patient's condition, medication order, and follow-up visit in OPD including Oncology Module

Pre-requisite for test	Test Validation	
1. The OPD module must be enabled in the system. 2. User must have valid login credentials with access rights to record and review OPD visits. 3. Patient master data must be available in the system. 4. Medication master list, allergy/adverse reaction records, and clinical diagnosis catalog must be pre-configured. 5. Oncology enhancement features must be enabled for follow-up status management.	Manual	
Steps to produce	Expected Outcome	Note/Deviation
Step 1. Navigate to the OPD module and select a registered patient for consultation. Step 2. Record the patient's condition including relevant medical history, current symptoms, physical examination findings, diagnostic results, and confirmed diagnosis. Step 3. Save the summary and verify that the system generates a concise overview of the patient's condition. Step 4. Navigate to the medication order section and prescribe medications by selecting drug name, dosage, frequency, and route of administration. Step 5. Record any allergy, adverse reaction, or drug interaction identified during prescription entry.	1. 1.A concise patient summary is generated, capturing medical history, symptoms, diagnosis, and examination/test findings. 2. Medication orders are recorded with complete details including dosage, frequency, route, and adverse reactions or drug interactions.	Select Yes/No

Steps to produce	Expected Outcome	Note/Deviation
Step 6. Review the system-generated medication history and validate its inclusion in the patient's OPD summary. Step 7. Navigate to the follow-up visit section and schedule a follow-up appointment including date, time, and clinician details. Step 8. Record follow-up consultation notes after the visit is completed. Step 9. Navigate to the oncology follow-up section and update follow-up status such as active surveillance, stable disease, no evidence of disease, relapse, referral, or long-term complications. Step 10. Capture and validate follow-up appointment details including planned visit dates and actual outcome (death, survivorship, or treatment discontinuation).	3. Medication history is integrated into the patient's OPD summary. 4. Follow-up visits are scheduled, documented, and retrievable in the system. 5. Follow-up consultation notes are recorded and linked to the patient's record. 6. Oncology-specific follow-up statuses are updated accurately in the patient's profile. 7. Planned visit dates and actual outcomes are captured and accessible in the system.	

COP1 : The system manages consultation services in OPD and IPD

COP1.c : The system creates order sets based on frequently prescribed medicines

Test Case: Verify creation and management of order sets for frequently prescribed medicines including Oncology Module

Pre-requisite for test	Test Validation	
- The medication database must be configured with drug names, dosage forms, routes, strengths, indications, contraindications, and side effects. - The system must have pre-established templates for common medical conditions or procedures. - User must have valid credentials with prescribing rights. - Oncology module must be enabled with chemotherapy drug pathways and brand selection options.	Manual	
Steps to produce	Expected Outcome	Note/Deviation
Step 1: Log in to the system and navigate to the Medication / Order Set Management section.	1. Pre-established order set templates	Select Yes/No

Steps to produce	Expected Outcome	Note/Deviation
Step 2: Create a new Order Set by entering a unique name, adding medicines, and defining dosage, frequency, route, duration, and any clinical notes or precautions. Step 3: Save the Order Set and reopen it to verify that all medication details and configurations are captured accurately. Step 4: Select an existing Order Set, modify medicine details or add/remove medicines, and verify that the system prompts for a Justification for Change. Step 5: Enter a valid justification, save the updates, and confirm that the changes-along with justification, user identity, timestamp-are recorded in the audit log or version history.	are available for common medical conditions or procedures. 2. Medical practitioners can personalise order sets by modifying or adding medicines. 3. Saved customised order sets are retrievable for future prescribing. 4. Predefined chemotherapy order sets are available with dosage, capping, and brand selection details	
For the Oncology requirement		
Step 6: Navigate to the Chemotherapy Order Set module and create a new predefined order set by selecting the disease group/organ or custom-defined pathway. Step 7: Add chemotherapy drugs to the order set, including brand selection, dosage details, and drug capping as per protocol. Step 8: Save the order set and verify that it appears correctly in the chemotherapy order set list. Step 9: Open the existing chemotherapy order set, modify drug details or selections, and verify that the system prompts for justification using a drop-down list. Step 10: Select a reason for modification, save the changes, and confirm that the justification is recorded in the audit log/version history.	5. Modifications to chemotherapy order sets require entry of justification. 6. Modified chemotherapy order sets are saved successfully with justification details recorded.	

COP.1 : The system manages consultation services in OPD and IPD

COP.1.g : The system has the capability to generate Computerized Provider Order Entry (CPOE) for radiological examinations.

Test Case : Verify CPOE functionality for radiological examinations including Oncology Radiation Therapy Enhancements

Pre-requisite for test	Test Validation
1. The radiology module must be enabled in the system. 2. The catalogue of radiology tests must be configured and available. 3. User must have valid credentials with medical practitioner access rights.	Manual

Pre-requisite for test		Test Validation	
4. Radiology scheduling and slot availability must be set up in the system. 5. Oncology radiation therapy module (Annexure B) must be enabled.			
Steps to produce		Expected Outcome	Note/Deviation
Step 1: Navigate to the patient record module and select a registered dummy patient. Step 2: Open the radiology test catalogue and select the required radiological test(s) along with the appropriate clinical indication. Step 3: Review the available schedules and slot timings for the selected radiology test. Step 4: Confirm patient preparation requirements, including fasting instructions, shifting time, and any special guidelines. Step 5: Place the radiology order and verify that it is successfully entered into the system. Step 6: Navigate to the Oncology radiation therapy module and Verify that the system allows adding oncology-related additional information for the selected test. Step 7: Record radiation therapy treatment details by selecting the required procedure type. Step 8: Choose the imaging modality such as CT, CT+MRI, or CT+PET. Step 9: Perform pretreatment verification and confirm that the system captures the verification details accurately. Step 10: View and mark the treatment checklist as defined by the healthcare facility. Step 11: Save the order and verify that it is accurately recorded in the patient's medical record.		1. Medical practitioners can place radiology test orders electronically using the CPOE system. 2. Radiology catalogue is accessible and tests can be selected with clear clinical indications. 3. Available schedules and slot timings are displayed and selectable for radiology tests. 4. Patient preparation instructions are recorded and linked to the order. 5. Radiology orders are successfully saved and visible in the patient's medical record. 6. Radiation therapy treatment details are captured including procedure type, imaging modality, and pretreatment verification. 7. Oncology-specific treatment checklists can be viewed and marked. 8. Completed oncology orders are saved and integrated into the patient's record.	Select Yes/No

COP.10 : The system supports patient services in remote settings.

COP.10.b : The system supports effective home care services/ palliative care.

Test Case : Verify management of effective home care and palliative care services including oncology enhancements.

Pre-requisite for test		Test Validation	
1. The user must have valid login credentials with access rights to home care and palliative care modules. 2. Patient records, including adult, paediatric, and cancer survivorship program participants, must exist in the system. 3. Home care service workflows, scheduling, and billing modules must be configured. 4. Oncology-specific home care modules, including patient profiling, adverse event tracking, remote monitoring, medication adherence, and teleconsultation capabilities, must be enabled. 5. Secure communication channels and document sharing mechanisms must be configured. 6. Diet, mental health, and advanced remote diagnostics modules must be active for home care management.		Manual	
Steps to produce		Expected Outcome	Note/Deviation
Step 1: Navigate to the Home Care Services module. Step 2: Create a new home care service booking by entering/selecting patient details, service type, and preferred schedule; save the booking. Step 3: Open the Home Care Dashboard and verify the booking appears with "Scheduled" status.if applicable Step 4: Select the appointment and update the status to In-Progress, then to Completed; ensure both updates reflect correctly. Step 5: Navigate to the Billing section, verify auto-generated charges, add/edit charges if required, and generate the invoice. Step 6: Confirm the invoice appears in the patient's billing history. Step 7: Verify that the system automatically sends a feedback form to the patient after service completion. Step 8: Submit the feedback and confirm it is recorded in the patient record and analytics dashboard.		1. The user is able to log in and access home care and palliative care modules successfully. 2. Home care services can be scheduled, assigned, and monitored efficiently. 3. Patient profiling and care details are accurately captured and retrievable. 4. Adverse events and remote monitoring data are recorded and tracked effectively.	Select Yes/No

Steps to produce	Expected Outcome	Note/Deviation
	5. Medication adherence and teleconsultation records are captured and monitored. 6. Billing for home care services is generated and maintained correctly. 7. Secure communication and document sharing are functional and reliable. 8. Oncology-specific modules capture diet, mental health, and advanced diagnostic data accurately. 9. Palliative care assessments and survivorship details are recorded as per defined guidelines. 10. All home care and palliative care data are stored securely, auditable, and accessible for authorized personnel.	

COP.12 : The system provides a Clinical Decision Support System.

COP.12.a : The Systems supports the Clinical Decision Support System (CDSS).

Test Case : Verify Clinical Decision Support System (CDSS) Functionality including Oncology Enhancements

Pre-requisite for test	Test Validation
1. The CDSS module must be enabled and configured in the system.	Manual

Pre-requisite for test		Test Validation	
- Access rights must be assigned to clinicians and authorized users. - Patient medical records including diagnosis, staging, pathology reports, and molecular reports must be available. - Medication database with dosage calculation rules (BSA, AUC, BMI, Height, Weight, Creatinine, Age, GFR) must be configured. - Integration with external CDSS systems, if applicable, must be active.		Manual	
Steps to produce		Expected Outcome	Note/Deviation
Step 1: Log in to the system with clinician credentials. Step 2: Navigate to the Clinical Decision Support System module. Step 3: Select a patient record and review available diagnosis, staging, and pathology data. Step 4: Trigger CDSS for diagnostic assistance and validate that evidence-based recommendations and guidelines are displayed. Step 5: Prescribe a medication and validate CDSS-generated alerts for contraindications, drug interactions, and dosage errors. Step 6: Initiate a treatment plan and verify CDSS suggestions for protocols based on patient diagnosis and staging. For the Oncology requirement Step 8: Review treatment protocol recommendations generated based on pathology reports, molecular findings, and host characteristics. Step 9: Validate that the system triggers an automated alert when diagnosis, staging, or pathology data is incomplete or inconsistent. Step 10: Enter patient demographic and clinical parameters (BSA, BMI, Age, Weight, Height, Creatinine, GFR). Step 11: Verify that the CDSS calculates and suggests appropriate drug dosage according to defined clinical rules. Step 12: Save and confirm that the recommendations and alerts are stored in the patient's medical record.		- CDSS provides evidence-based recommendations for diagnosis and treatment planning. - System generates alerts for contraindications, drug interactions, and prescription errors. - CDSS suggests appropriate treatment protocols based on diagnosis, staging, and pathology reports. - Oncology-specific treatment recommendations are generated considering molecular pathology and host characteristics. - Automated alerts are triggered when required data is incomplete or incongruent. - Dosage suggestions are accurately calculated based	Select Yes/No

Steps to produce	Expected Outcome	Note/Deviation
	on clinical parameters (BSA, AUC, BMI, Height, Weight, Creatinine, Age, GFR). 7. All CDSS recommendations and alerts are recorded in the patient's electronic medical record.	

COP.13 : The system has the capability to create care plans.

COP.13.a : The system has the capability to create customized care plans based on current standards of practice

Test Case : Verify the creation of customized care plans for disease-specific and oncology patients

Pre-requisite for test	Test Validation	
1. The user must have valid login credentials with access rights to care plan creation and management. 2. Disease-specific clinical guidelines (e.g., asthma, diabetes, COPD, oncology) should be pre-configured in the system. 3. Oncology-specific protocols (e.g., NCG guidelines, clinician-defined protocols) must be available in the system. 4. Radiotherapy and chemotherapy modules should be enabled in the system. 5. Patient records with relevant demographic and clinical details must be available in the system.	Manual	
Steps to produce	Expected Outcome	Note/Deviation
Step 1: Log in to the system with valid user credentials. Step 2. Navigate to the Care Plan Management module. Step 3. Select a patient record from the patient list. Step 4. Initiate the creation of a new customized care plan. Step 5. Select a disease-specific template (e.g., asthma, diabetes, COPD) or oncology-specific protocol (e.g., NCG guideline, clinician-defined protocol).	1. The system allows successful login and access to the Care Plan Management module. 2. The user is able to initiate and create a new customized care plan for a selected patient.	Select Yes/No

Steps to produce	Expected Outcome	Note/Deviation
Step 6. Configure care plan elements including assessment and evaluation, goal setting, interventions, monitoring and adjustments, documentation, and communication. For the Oncology requirement Step 7. For oncology patients, configure additional elements including comorbidity management, cancer-specific care, pain management, follow-up plans, survivorship plans, or palliative care as required. Step 8. Access the chemotherapy module to define or modify chemotherapy sequences, cycle intervals, treatment types, reasons for changes, outcomes, and user remarks. Step 9. Access the radiotherapy module to configure requisition details such as session counts, cycle intervals, protocols, site of incidence, region of interest, procedure dates, and diagnostic findings (e.g., PET-CT/CT). Step 10. Save and finalize the customized care plan. Step 11. Review the complete audit trail to ensure all actions are logged as per facility requirements. Step 12. Reopen the saved care plan to verify the ability to view, update, or modify details.	3. Disease-specific care plans are generated as per configured guidelines and include assessment, interventions, monitoring, and documentation. 4. Oncology care plans are successfully created, tailored to organ, disease group, stage, and tumour site, and include comorbidity management, cancer care, pain management, and follow-up or survivorship plans. 5. The system enables configuration and modification of chemotherapy plans, including treatment sequences, reasons for changes, and outcomes. 6. The system enables configuration of radiotherapy requisition details in alignment with institutional protocols. 7. All care plan activities are logged and viewable in the audit trail.	

Steps to produce	Expected Outcome	Note/Deviation
	8. The system allows users to view, update, and modify care plans after creation without data loss or errors.	Select Yes/No

MOM.3: The system supports the safe administration of medications

MOM.3.b: The system has the capability of maintaining an electronic medication administration record (eMAR)

Test Case : Verify the maintenance of electronic medication administration record (eMAR) with oncology enhancements

Pre-requisite for test	Test Validation
1. The user must have valid login credentials with access rights to medication administration modules. 2. Patient records with prescribed medications must exist in the system. 3. Medication templates must be pre-configured in the system, including dosage, route, date/time, and administering personnel details. 4. Oncology-specific enhancements (chemotherapy and radiotherapy orders, dilution details, premedication, and post-chemotherapy medications) must be enabled. 5. Defined drop-down lists for medication status (completed, canceled, modified with reasons) should be available in the system.	Manual

Steps to produce	Expected Outcome	Note/Deviation
Step 1: Log in to the system with valid user credentials. Step 2: Navigate to the eMAR module. Step 3: Select a patient record from the patient list. Step 4: Record a new medication administration using the prescribed template, entering dosage, route of administration, date and time, and administering personnel details. Step 5: Record a medication administered based on a verbal order and capture the verification details for high-risk medication. For the Oncology requirement Step 6: For oncology patients, enter details of parenteral drug dilution, diluent, volume, and infusion rate.	1. The system allows successful login and access to the eMAR module. 2. The user is able to record medication administration using the predefined template, including dosage, route, date, time, and administering personnel details.	Select Yes/No

Steps to produce	Expected Outcome	Note/Deviation
Step 7. Record premedication details along with post-chemotherapy medication details such as duration, frequency, and name. Step 8. Access chemotherapy orders for the patient and review them date-wise, verifying the status as completed, canceled, or modified, along with the reason selected from the drop-down list. Step 9. Capture planned and actual chemotherapy session details, including date, time, and type (e.g., adjuvant). Step 10. Capture planned and actual radiotherapy session details, including date, time, and type. Step 11. Save the medication administration record. Step 12. Reopen the eMAR to validate that all recorded details are accurately stored, viewable, and auditable.	3. The system records and distinguishes verbal order-based medication entries with verification for high-risk medications. 4. Oncology-specific details such as dilution, diluent, volume, rate, premedication, and post-chemotherapy medications are successfully recorded. 5. Date-wise chemotherapy orders are displayed with accurate statuses (completed, canceled, modified) and associated reasons from the defined drop-down list. 6. Planned and actual chemotherapy and radiotherapy session details (date, time, type) are successfully captured. 7. All medication administration entries are stored in the eMAR and are viewable, editable, and auditable as per facility requirements.	

IMS.1: The system supports healthcare data and interoperability standards for patient, clinical, administrative information to ensure continuity of care, including ABDM

IMS.1.e : The system shall support ICD 10/11 or SNOMED CT covering clinic terminologies for diagnosis, morbidity, and mortality data accurately.

Test Case : Verify system support for ICD 10/11, SNOMED CT, ICD-O/ICD-O3, and CTCAE v5.0 coding functionality

Pre-requisite for test		Test Validation	
1. The user must have valid login credentials with access rights to diagnosis and coding modules. 2. ICD 10/11, SNOMED CT, ICD-O/ICD-O3, and CTCAE v5.0 code sets must be configured and available in the system. 3. The system must be updated to support upload, upgrade, deprecation, and version management of medical coding standards. 4. Patient records must exist in the system for assigning diagnosis and treatment-related codes.		Manual	
Steps to produce		Expected Outcome	Note/Deviation
Step 1: Log in to the system with valid user credentials. Step 2: Navigate to the diagnosis and coding module. Step 3: Search for a diagnosis using clinical terminology and verify system recommendations for relevant ICD 10/11 and SNOMED CT codes. Step 4: Select and assign the appropriate ICD 10/11 or SNOMED CT code to the patient record. Step 5: Verify that the system maintains the coding version and supports retrieval of historical versions. Step 6: Upload or upgrade a new version of ICD or SNOMED CT codes into the system. Step 7: Validate that deprecated codes are clearly marked and retained for historical accuracy. For oncology requirements Step 8: For oncology cases, record ICD-O/ICD-O3 codes to capture morphology and grade of the disease. Step 9: Capture CTCAE v5.0 codes for recording treatment-related toxicities, including grade and category. Step 10: Save and review the coded patient record to ensure accurate storage and retrieval of all assigned codes.		1. The system allows successful login and access to the coding module. 2. Relevant ICD 10/11 or SNOMED CT codes are recommended based on clinical terminology search. 3. The user is able to assign appropriate codes to patient records. 4. The system maintains and displays coding versions, with the ability to upload, upgrade, and deprecate codes while retaining historical data. 5. Deprecated codes are clearly identified and not used for new entries, but remain visible for past records.	Select Yes/No

Steps to produce	Expected Outcome	Note/Deviation
	6. Oncology-specific codes (ICD-O/ICD-O3) for morphology and grade are successfully recorded. 7. CTCAE v5.0 codes are captured accurately for treatment-related toxicities with grade and category. 8. The coded patient record is stored, retrievable, and auditable as per clinical and institutional requirements.	

IMS.2: The system has the capability to support NABH defined key performance indicators and analytical dashboards.

IMS.2.a : The system electronically computes and publishes Key Performance Indicators (KPIs) per NABH accreditation standards for hospitals and healthcare organizations

Test Case : Verify computation and publication of NABH-compliant KPIs including oncology-specific data.

Pre-requisite for test		Test Validation	
1. The user must have valid login credentials with access rights to the KPI computation module. 2. Relevant patient, administrative, and hospital operational data must be available in the system. 3. NABH accreditation standards and KPI definitions must be configured in the system (Annexure 8 / Annexure F for oncology-specific KPIs). 4. KPI export formats (JSON, CSV, XML, XLS, PDF) must be pre-configured and accessible. 5. Oncology module must be enabled for capturing cancer registry-related data.		Manual	
Steps to produce	Expected Outcome	Note/Deviation	
Step 1. Log in to the system with valid credentials. Step 2. Navigate to the KPI computation module. Step 3. Select the desired reporting period by entering start and end dates.	1. The user is able to log in and access the KPI computation module successfully.	Select Yes/No	

Steps to produce		Expected Outcome	Note/Deviation
Step 4.	Initiate the KPI computation process.	- The system allows selection of custom reporting periods and computes KPIs accordingly. - KPIs are computed accurately based on NABH accreditation standards. - Computed KPI results match reference or expected data values. - KPI data can be exported successfully in all configured formats (JSON, CSV, XML, XLS, PDF). - Oncology-specific KPIs for cancer registries are captured, computed, and exported correctly. - Published KPIs are accessible to authorized users and suitable for further analysis. - The system maintains data integrity and traceability for all KPI computations and exports.	
Step 5.	Verify that the system accurately computes KPIs based on NABH accreditation standards.		
Step 6.	Access computed KPI results and validate the accuracy against sample or reference data.		
Step 7.	Export the KPI data in multiple formats (JSON, CSV, XML, XLS, PDF) and verify successful export.		
Step 8.	For oncology-specific KPIs, capture and verify required information for cancer registries.		
Step 9.	Validate that oncology KPIs are correctly formatted and exportable as per defined NABH standards.		
Step 10.	Review published KPIs to ensure they are accessible to authorized end-users for analysis.		

MDC.1: The system creates a Multi-Disciplinary Tumor Board workflow.

MDC.1.a: The system enables patient selection, tumor board ID assignment, MDTB review scheduling, and automated clinician notifications.

Test Case : Verify that the system allows a treating medical practitioner to select a patient, assign a tumor board ID, schedule a meeting, add practitioners, and send notifications.

Pre-requisite for test		Test Validation	
1. Healthcare provider is logged in to the system using valid login credentials. 2. Dummy patient records are available in the system with complete demographic and clinical details. 3. Configurable clinical criteria (e.g., confirmed/possible diagnosis, TNM stage, organ system) are predefined in the system. 4. Clinicians required for the MDTB are registered in the system with valid contact details and calendar integration enabled.		Manual	
Steps to produce		Expected Outcome	Note/Deviation
Step 1. Navigate to the section related to MDTB module. Step 2. Search and select a dummy patient. Step 3. Create a new Tumor Board Case for the dummy patient. Step 4. Verify that the system assigns a unique Tumor Board ID linked to the patient's unique ID. Step 5. Enter clinical criteria for MDTB scheduling (diagnosis, TNM stage, organ system). Step 6. Set the date and time for the tumor board meeting. Step 7. Select the required medical practitioners (e.g., oncologist, radiologist, pathologist). Step 8. Save and finalize the MDTB meeting. Step 9. Check that appointment notifications are sent to all selected practitioners. Step 10. Verify that each selected practitioner's calendar is blocked for the scheduled time.		1. Medical practitioner can successfully select patient for MDTB. 2. System is able to generate a unique Tumor Board ID and correctly link with the patient's unique ID. 3. MDTB meeting is scheduled with the given clinical criteria. 4. Notifications are sent to all selected stackholders. 5. Medical Practitioners calendars show the scheduled meeting blocked.	Select Yes/No

MDC.1: The system creates a Multi-Disciplinary Tumor Board workflow.

MDC.1.b: The system displays the list of patients scheduled for MDTB meetings, allowing the medical practitioner to select and view patient-specific details on a single screen.

Test Case : Verify that the system can generate centralized patient list and integrated data view for tumor board meetings.

Pre-requisite for test	Test Validation
1. Healthcare provider is logged in to the system using valid login credentials. 2. Dummy patients are scheduled for MDTB meetings with confirmed dates and times.	Manual

Pre-requisite for test	Test Validation	
3. Clinical data, including case notes, lab results, procedure details, radiology/pathology images, and supporting documents, are available in EMR, PACS, LIS, and RIS.	Manual	
Steps to produce	Expected Outcome	Note/Deviation
Step 1. Navigate to the section related to MDTB module and open the patient schedule view. Step 2. Verify the list of patients scheduled for upcoming MDTB meetings is displayed. Step 3. Apply filters such as date, specialty, or disease category to refine the patient list. Step 4. Select a dummy patient from the displayed list. Step 5. View the patient-specific details consolidated on a single screen, including case notes, lab results, procedure details, radiology/ pathology images, and supporting documents. Step 6. Check and confirm that the accurate data is seamlessly pulled from EMR, PACS, LIS, and RIS without errors or delays.	1. The system successfully displays the complete list of patients scheduled for MDTB meetings. 2. Filters by date, specialty, or disease category refine the patient list accurately. 3. Selection of a patient opens a consolidated view of patient-specific details on a single screen. 4. Clinical data from EMR, PACS, LIS, and RIS are displayed seamlessly and comprehensively.	Select Yes/No

MDC.1: The system creates a Multi-Disciplinary Tumor Board workflow.

MDC.1.c : The system records the attendance and details of medical practitioners who attend the MDTB meeting, and documents them as per the department.

Test Case : Verify that the system allows authorized users to view assigned medical practitioners for MDTB meetings, record attendance details (name, designation, specialty, and role), and send automated calendar reminders to participants.

Pre-requisite for test	Test Validation
1. Healthcare provider is logged in to the system using valid login credentials. 2. The MDTB meeting is already scheduled with assigned medical practitioners. 3. Medical practitioners' profiles are available in the system, including name, designation, specialty, and department. 4. Calendar reminders for MDTB meetings are enabled in the system.	Manual

Steps to produce	Expected Outcome	Note/Deviation
Step 1: Login to the system using authorized user to access MDTB meeting schedule. Step 2: Navigate to the “MDTB Meetings” module in the system. Step 3: Select an upcoming MDTB meeting from the list. Step 4: Verify that the list of assigned medical practitioners for the selected meeting is displayed. Step 5: During/after the meeting, navigate to the “Attendance” section. Step 6: Record attendance by marking practitioners as present or absent. Step 7: Verify that the system captures details of attending practitioners and the presenter including: Name Designation Specialty Role in proceedings Step 8: Verify that attendance records are saved and retrievable. Step 9: Check whether automated calendar reminders are sent to assigned practitioners before the scheduled meeting. Step 10: Confirm that reminders include meeting details (date, time, agenda) and are delivered on time.	1. The system allows access to the scheduled MDTB meeting details to the authorized users. 2. The list of assigned medical practitioners for MDTB meeting is displayed accurately. 3. Attendance is successfully recorded for practitioners who participated in the meeting. 4. Practitioner details, including name, designation, specialty, and department, are correctly documented. 5. Attendance records are saved and categorized by department as per MDT documentation standards.	Select Yes/No

MDC.1: The system creates a Multi-Disciplinary Tumor Board workflow.

MDC.1.d: The system has the capability to document a summary of the tumor board discussion related to the diagnoses and treatment recommendations.

Test Case: Verify Documentation of MDTB Discussion summary on diagnoses and treatment recommendations

Pre-requisite for test	Test Validation
1. Healthcare provider or administrative staff is logged into the system using valid login credentials. 2. An MDTB meeting is scheduled and conducted with medical practitioners' participation. 3. Individual inputs related to diagnosis, treatment recommendations, and follow-ups are recorded in the system.	Manual

Pre-requisite for test		Test Validation	
4. The MDTB summary template (annexure D) is configured and available in the system.		Manual	
Steps to produce		Expected Outcome	Note/Deviation
Step 1. Navigate to the MDTB module and open the completed meeting record. Step 2. Enter individual inputs provided by practitioners related to diagnosis, treatment recommendations, and follow-ups. Step 3. Generate the summary of the MDTB discussion using the standardized template provided in annexure D. Step 4. Review the summary to ensure it consolidates multidisciplinary inputs in a structured format. Step 5. Save and finalize the MDTB summary record. Step 6. Verify that the summary is linked to the patient's medical record for future reference.		1. The system allows access to the completed MDTB meeting record. 2. Individual inputs related to diagnosis, treatment recommendations, and follow-ups are successfully entered. 3. The system generates a summary of the MDTB discussion using the standardized template. 4. The summary consolidates multidisciplinary inputs in a clear and structured format. 5. The finalized MDTB summary is saved without errors.	Select Yes/No

MDC.2 : The system can capture details related to research and trials.

MDC.2.a: The system has the provision to capture consent, label patients enrolled under research trials, and view the list of research trials conducted.

Test Case : Verify capture of consent, patient enrollment, and viewing of research trials

Pre-requisite for test		Test Validation	
1. Healthcare provider or administrative staff is logged in to the system using valid login credentials. 2. Patient records are available in the system with complete demographic and clinical details.		Manual	

Pre-requisite for test		Test Validation	
3. Research trials are configured in the system with trial name and trial ID. 4. The consent template (annexure – sample consent template) is available in the system..		Manual	
Steps to produce		Expected Outcome	Note/Deviation
Step 1. Navigate to the research trial module. Step 2. Select a patient from the patient list to evaluate for a research trial. Step 3. Capture screening details by marking “Screened for trial – Yes/No” in the system. Step 4. Record eligibility status by marking “Eligible for trial – Yes/No.” Step 5. Capture consent status by marking “Consent given – Yes/No.” Step 6. If consent is given, upload the signed consent form using the reference of annexure template. Step 7. Enter the trial details, including Trial Name and Trial ID. Step 8. Save the patient enrollment details in the system. Step 9. Navigate to the research trial list view and verify the list of patients enrolled in ongoing and past trials.		1. The system allows access to the research trial module. 2. Screening status of the patient for the trial is recorded successfully. 3. Eligibility details are captured accurately. 4. Consent status is documented, and the consent form is uploaded (if applicable). 5. Trial Name and Trial ID are recorded correctly. 6. The patient is labeled as enrolled under the research trial. 7. The list of patients enrolled in ongoing and past research trials is displayed accurately.	Select Yes/No

MDC.2. : The system can capture details related to research and trials.

MDC.2.b: The system supports the end-to-end management of clinical and research trials, including randomization and enrollment, visit tracking, investigational product management, adverse event monitoring, and close-out and archival.

Test Case : Verify end-to-end functionality of Clinical and Research Trial Management module.

Pre-requisite for test	Test Validation
1. Authorized medical practitioner must have access to the Clinical Trial Management module. 2. Trial protocol, patient master data, and consent records must be available in the system.	Manual

Pre-requisite for test		Test Validation	
3. Investigational product (drug/device) inventory must be configured and available in the system. 4. Role-based access control must be enabled for users managing clinical and research trials.		Manual	
Steps to produce		Expected Outcome	Note/Deviation
Step 1. Navigate to the section related to Clinical Trial Management module and initiate a new trial setup. Step 2. Configure trial protocol details, including study objectives, phases, and ethics committee approval references. Step 3. Execute randomization logic by assigning enrolled participants to appropriate trial arms and verify system logs for participant allocation. Step 4. Schedule and configure study-specific visits and procedures, ensuring visit timelines and adherence tracking are properly defined. Step 5. Record and track visit completion for each participant, verifying that the system updates visit status accordingly. Step 6. Access the Investigational Product Management section and record receipt, batch numbers, and dispensing details for investigational products. Step 7. Verify that the system maintains accountability records and dispensing logs linked to participant identifiers. Step 8. Document adverse events in the Adverse Event Monitoring section, classifying them by type, severity, and causality. Step 9. Confirm that each recorded adverse event is correctly linked to the relevant trial protocol and participant record. Step 10. Proceed to Close-Out & Archival and close the trial with a final status update. Step 11. Verify that all associated documents, including approvals, monitoring reports, and event logs, are archived systematically. Step 12. Check that archived records are retrievable and accessible only to authorized roles for audit or regulatory review.		1. The system successfully enables creation and configuration of clinical or research trials. 2. Participant randomization and enrollment are accurately recorded with traceable logs. 3. Study visits are scheduled, tracked, and updated based on adherence and completion. 4. Investigational product management captures complete batch, dispensing, and accountability details. 5. Adverse events are documented, classified, and linked correctly to respective protocols and participants. 6. The trial closure and archival processes complete successfully with all related records securely stored. 7. Archived data is retrievable and accessible only to authorized users, ensuring regulatory compliance and data integrity.	Select Yes/No

MDC.3 : The system captures details related to bone marrow transplantation.

MDC.3.a : The system has the provision to enter donor and recipient details in defined templates for bone marrow transplant.

Test Case : Verify Entry of Donor and Recipient Details for Bone Marrow Transplant Using Defined Templates

Pre-requisite for test		Test Validation	
1. Healthcare provider or administrative staff is logged in to the system using valid login credentials. 2. Defined templates for both autologous and allogeneic transplants are configured in the system. 3. Patient and donor records are available in the system with complete demographic and clinical details. 4. Consent form and transplant committee approval templates (annexure) are available in the system.		Manual	
Steps to produce		Expected Outcome	Note/Deviation
Step 1.	Navigate to the bone marrow transplant module.	1. The system allows access to the bone marrow transplant module. 2. The defined templates for both autologous and allogeneic transplants are displayed correctly. 3. Donor and recipient details are successfully entered as per the selected transplant type. 4. Supporting documents, including consent forms and committee approvals, are uploaded and linked to the record.	Select Yes/No
Step 2.	Select the type of transplant (Autologous or Allogeneic).		
Step 3.	For an Autologous transplant, enter the following details in the defined template: indication, disease status, pre-transplant work-up, stem cell mobilization agents, conditioning therapy with dates, date of stem cell infusion, stem cell dose, date of neutrophil engraftment, date of platelet engraftment, and any significant complications.		
Step 4.	For an Allogeneic transplant, enter the following details in the defined template: indication, disease status, pre-transplant work-up, type of donor, donor work-up details, conditioning therapy with dates, date of stem cell infusion, stem cell dose, date of neutrophil engraftment, date of platelet engraftment, GVHD prophylaxis, acute GVHD details, chronic GVHD, treatment of GVHD, and any significant complications.		
Step 5.	Upload supporting documents such as the consent form, psychiatric evaluation, and doctor's recommendation, if applicable.		
Step 6.	Process and record transplant committee approval for the procedure.		

Steps to produce	Expected Outcome	Note/Deviation
Step 7. Save the donor and recipient details in the system. Step 8. Share the relevant reports with authorized agencies through the system's reporting feature.	5. The system processes and records internal approvals for the transplantation procedure. 6. Donor and recipient details are saved without errors in the system. 7. Reports are shared accurately with authorized agencies as per defined access rights.	



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